

T_1 acts as -1 on $B^{(\lambda)}V_\lambda$ in the deep stable regime:
a structural theorem dual to the T_ℓ -eigenvalue

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Abstract

The structural-skeleton paper of this evening (`2026-05-13-evening-T-ell-scalar.tex`) proved a universal fact: T_ℓ acts as the scalar q on $B^{(\lambda)}V_\lambda$ for every λ with $\lambda_\ell = 1$. The morning refutation paper (`2026-05-13-decoration-iso-failed.tex`) observed empirically that T_1 acts as -1 on B for certain test shapes but *not* universally — the action depends on the shape, with a clear threshold in the trailing-1-row count.

This paper isolates the structural condition. We prove:

Theorem. Let λ be a partition with $\lambda_\ell = 1$. Write $\lambda = (\hat{\lambda}, 1^q)$ where $\hat{\lambda}$ has all rows of length ≥ 2 . Assume:

- (a) The SRD-style reduction holds at λ and at every shape encountered in the recursive descent of λ .
- (b) $q \geq q_0(\hat{\lambda}) := |\hat{\lambda}| - 2\ell(\hat{\lambda}) + 2$.

Then

$$B_{\ell+1}^{(\lambda)}V_\lambda \subseteq E_1^-|_{V_\lambda}.$$

The proof is by induction on $|\hat{\lambda}^{(\geq 2)}|$. The inductive step combines three ingredients: (i) T_1 -equivariance of the corner-pair branching injection Φ_i^λ (an index-gap ≥ 2 statement), (ii) preservation of E_1^- by $\mathcal{O}_\lambda = (T_\ell+1)\cdots(T_{n-2}+1)$ when $\ell \geq 3$ (also index-gap ≥ 2), and (iii) the SRD-style reduction at the level of V_λ . The threshold $q_0(\hat{\lambda})$ is precisely the condition that $\ell \geq 3$ holds at every step of the descent. The threshold q_0 matches the empirical sharp threshold on every test shape examined.

Combined with the evening structural-skeleton fact, this gives the two scalar actions on B in the deep stable regime: T_ℓ acts as $+q$ universally; T_1 acts as -1 once $q \geq q_0(\hat{\lambda})$.

1 Setup and notation

Notation follows the May-13 multi-corner paper and the $r=1$ inductive reduction paper (`2026-05-13-r1-inductive` `2026-05-13-r-additivity-and-syt-formula.tex`). $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ obeying $(T_i - q)(T_i + 1) = 0$ and the standard braid relations. V_λ is the seminormal Specht module in asymmetric Murphy normalisation ($b' = 1$). For $j \in \{1, \dots, n-1\}$, $E_j^\pm := E_j^\pm(V_\lambda)$ denote the q - and (-1) -eigenspaces of T_j . For $1 \leq k \leq n$,

$$R'_k := (T_{k-1}+1)(T_{k-2}+1)\cdots(T_1+1), \quad B_{j_0}^{(\lambda)}V_\lambda := R'_{j_0}\cdots R'_nV_\lambda$$

with $j_0 := \ell + 1$, $\ell := \ell(\lambda)$.

Partition data. Write $\lambda = (\hat{\lambda}, 1^q)$ where $\hat{\lambda}$ is the part with rows of length ≥ 2 and q is the number of trailing 1-rows. The *length-preserving corners* of λ are the removable corners in column ≥ 2 , i.e., $(\rho_i, \hat{\lambda}_{\rho_i})$ where ρ_i is a row of $\hat{\lambda}$ that is a corner of $\hat{\lambda}$ (no row below it of equal length). List them as $c_1, \dots, c_r$ in top-to-bottom order. The *column-1 bottom corner* is $c_* := (\ell, 1)$ when $q \geq 1$. Throughout this paper we assume $q \geq 1$, so c_* exists.

Decoration shape. $\hat{\lambda}^{(\geq 2)} := \hat{\lambda}$ with column 1 deleted. Then $|\hat{\lambda}^{(\geq 2)}| = |\hat{\lambda}| - \ell(\hat{\lambda})$. The threshold function of this paper is

$$q_0(\hat{\lambda}) := |\hat{\lambda}| - 2\ell(\hat{\lambda}) + 2 = |\hat{\lambda}^{(\geq 2)}| - \ell(\hat{\lambda}) + 2. \quad (1)$$

Corner-pair branching injections. For each length-preserving corner c_i , set $\mu_i := \lambda \setminus \{c_i, c_*\}$, a partition of $n - 2$ with $\ell(\mu_i) = \ell - 1$. The branching injection $\Phi_i^\lambda : V_{\mu_i} \hookrightarrow V_\lambda$ sends a seminormal basis vector $v_{T'}^{(\mu_i)}$ to the unique $+q$ -eigenvector of T_{n-1} on the 2-block $\langle v_{T_A^i}, v_{T_B^i} \rangle$ of V_λ , where T_A^i, T_B^i are the two extensions of T' placing $\{n-1, n\}$ at $\{c_i, c_*\}$ in the two orders. The non-degeneracy of the 2-block (c_i in column ≥ 2 non-bottom row, c_* in column 1 bottom row) ensures Φ_i^λ is well-defined and $\langle T_1, \dots, T_{n-3} \rangle$ -equivariant (see **LB-closed** §1 or the r=1 paper Lemma 3.4 for details).

Outer chain.

$$\mathcal{O}_\lambda := (T_\ell + 1)(T_{\ell+1} + 1) \cdots (T_{n-2} + 1).$$

SRD-style reduction. We say the *SRD-style reduction holds at λ* if

$$B_{\ell+1}^{(\lambda)} V_\lambda = \mathcal{O}_\lambda \left[\sum_{i=1}^r \Phi_i^\lambda (B_{j_0(\mu_i)}^{(\mu_i)} V_{\mu_i}) \right]. \quad (2)$$

This is the form of the SRD reduction when all “other” pieces (same-row, pure-length-pres) vanish; it has been proved unconditionally for various shape families and is the central hypothesis of the present paper. We say λ is in the *fully stable regime* if (2) holds at λ and at every shape encountered in the recursive descent $\lambda \rightsquigarrow \mu_i \rightsquigarrow \mu_{i,j} \rightsquigarrow \cdots$ down to a base case.

Descent base case. The recursive descent terminates at shapes λ' with no length-preserving corners ($r = 0$), equivalently $\lambda' = (1^k)$ for some $k \geq 1$. At a base case, (2) is vacuous (no μ_i 's).

2 Inductive step

The inductive step is short and rests on two index-gap commutations and the T_1 -equivariance of Φ_i^λ .

Lemma 1 (Φ_i^λ is T_1 -equivariant for $|\mu_i| \geq 4$). *For $|\mu_i| \geq 4$ (equivalently $n - 2 \geq 4$, i.e., $n \geq 6$),*

$$T_1 \cdot \Phi_i^\lambda(v) = \Phi_i^\lambda(T_1 v) \quad \text{for every } v \in V_{\mu_i}.$$

In particular, $\Phi_i^\lambda(E_1^-|_{V_{\mu_i}}) \subseteq E_1^-|_{V_\lambda}$.

Proof. Φ_i^λ is $\langle T_1, \dots, T_{n-3} \rangle$ -equivariant by construction (r=1 paper Lemma 3.4, or **LB-closed** §1). For the operator T_1 to lie inside this range we need $1 \leq n - 3$, i.e., $n \geq 4$, equivalently $|\mu_i| = n - 2 \geq 2$. The slightly stronger hypothesis $n \geq 6$ (i.e., $|\mu_i| \geq 4$) is used below in the chain of inductive applications but is not required at this lemma alone. The E_1^- -preservation is immediate: if $T_1 v = -v$, then $T_1 \Phi_i^\lambda(v) = \Phi_i^\lambda(T_1 v) = -\Phi_i^\lambda(v)$. $\square$

Lemma 2 ($\mathcal{O}_\lambda$ preserves $E_1^-|_{V_\lambda}$ for $\ell \geq 3$). *If $\ell(\lambda) \geq 3$, the operator $\mathcal{O}_\lambda$ commutes with T_1 as an element of $H_q(S_n)$. In particular, $\mathcal{O}_\lambda(E_1^-|_{V_\lambda}) \subseteq E_1^-|_{V_\lambda}$.*

Proof. $\mathcal{O}_\lambda = (T_{\ell+1})(T_{\ell+1+1}) \cdots (T_{n-2+1})$. Each factor is (T_j+1) for $j \in \{\ell, \dots, n-2\}$. Since $\ell \geq 3$, every such j satisfies $j \geq 3$, hence $|j-1| \geq 2$, so T_j commutes with T_1 inside $H_q(S_n)$ (the commuting Hecke relation for non-adjacent generators). Therefore (T_j+1) commutes with T_1 and preserves any T_1 -eigenspace. The chain product preserves E_1^- . $\square$

Theorem 3 (Inductive step). *Let λ be a partition with $\lambda_\ell = 1$ and $\ell(\lambda) \geq 3$. Assume the SRD-style reduction (2) holds at λ . Suppose for each $i \in \{1, \dots, r\}$, $B_{j_0(\mu_i)}^{(\mu_i)} V_{\mu_i} \subseteq E_1^-|_{V_{\mu_i}}$. Then*

$$B_{\ell+1}^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda}.$$

Proof. By hypothesis, $B^{(\mu_i)} V_{\mu_i} \subseteq E_1^-|_{V_{\mu_i}}$. For $|\mu_i| \geq 2$, Φ_i^λ is T_1 -equivariant (Lemma 1), so

$$\Phi_i^\lambda(B^{(\mu_i)} V_{\mu_i}) \subseteq \Phi_i^\lambda(E_1^-|_{V_{\mu_i}}) \subseteq E_1^-|_{V_\lambda}.$$

Summing over i , $\sum_i \Phi_i^\lambda(B^{(\mu_i)} V_{\mu_i}) \subseteq E_1^-|_{V_\lambda}$. By Lemma 2 ($\ell \geq 3$), $\mathcal{O}_\lambda$ preserves $E_1^-|_{V_\lambda}$, so

$$B^{(\lambda)} V_\lambda = \mathcal{O}_\lambda \left[\sum_i \Phi_i^\lambda(B^{(\mu_i)}) \right] \subseteq \mathcal{O}_\lambda(E_1^-|_{V_\lambda}) \subseteq E_1^-|_{V_\lambda}.$$

The equality uses (2).

Edge case $|\mu_i| = 1$ or 2: this occurs only when $n \leq 4$, hence $\ell(\lambda) \leq 2$, ruled out by hypothesis $\ell \geq 3$. For $|\mu_i| = 3$, i.e., $n = 5$, the operator T_1 lies in $\langle T_1, T_2 \rangle = \langle T_1, \dots, T_{n-3} \rangle$, so Φ_i^λ is T_1 -equivariant by the same proof. No issue. $\square$

3 Base case

Lemma 4 (Column shape base case). *For $\lambda = (1^k)$ with $k \geq 1$:*

$$B_{\ell+1}^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda}.$$

Proof. $V_{(1^k)}$ is the 1-dimensional sign representation of $H_q(S_k)$. Every T_j acts as -1 , so $V_\lambda = E_j^-$ for every j , in particular for $j = 1$. Hence any subspace — including $B^{(\lambda)} V_\lambda$ — is contained in E_1^- . (When $k \geq 2$ this is meaningful; when $k = 1$, V_λ is 1-dimensional and the T_1 -eigenspace question is vacuous.) $\square$

4 Threshold propagation

We verify that the threshold $q_0(\hat{\lambda}) = |\hat{\lambda}| - 2\ell(\hat{\lambda}) + 2$ controls both (i) the inductive lemma hypothesis $\ell \geq 3$ at λ and (ii) the propagation λ -threshold $\Rightarrow \mu_i$ -threshold.

Lemma 5 (Threshold $\Rightarrow \ell \geq 3$). *Let $\lambda = (\hat{\lambda}, 1^q)$ with $\hat{\lambda} \neq \emptyset$ (so $\ell(\hat{\lambda}) \geq 1$). If $q \geq q_0(\hat{\lambda}) = |\hat{\lambda}| - 2\ell(\hat{\lambda}) + 2$, then $\ell(\lambda) \geq |\hat{\lambda}^{(\geq 2)}| + 2 \geq 3$.*

Proof. $\ell(\lambda) = \ell(\hat{\lambda}) + q$. The threshold gives $q \geq |\hat{\lambda}| - 2\ell(\hat{\lambda}) + 2$, so $\ell(\lambda) = \ell(\hat{\lambda}) + q \geq |\hat{\lambda}| - \ell(\hat{\lambda}) + 2 = |\hat{\lambda}^{(\geq 2)}| + 2$. Since $\hat{\lambda} \neq \emptyset$ implies $|\hat{\lambda}^{(\geq 2)}| \geq 1$ (every row of $\hat{\lambda}$ has length ≥ 2 , contributing ≥ 1 to $\hat{\lambda}^{(\geq 2)}$), we get $\ell(\lambda) \geq 3$.

Boundary case $\hat{\lambda} = \emptyset$: then $\lambda = (1^q)$, which is the base case (Lemma 4), not handled by the inductive step. $\square$

Lemma 6 (Threshold propagates to each μ_i). *Let $\lambda = (\hat{\lambda}, 1^q)$ with $\hat{\lambda} \neq \emptyset$ and $q \geq q_0(\hat{\lambda})$. Then for every length-preserving corner c_i of λ (i.e., corner of $\hat{\lambda}$),*

$$q(\mu_i) \geq q_0(\hat{\mu}_i).$$

Proof. Let $c_i = (\rho_i, \hat{\lambda}_{\rho_i})$ be a length-preserving corner. We split into two cases based on whether $\rho_i = \ell(\hat{\lambda})$ and $\hat{\lambda}_{\rho_i} = 2$.

Case 1: $\rho_i < \ell(\hat{\lambda})$, or $\rho_i = \ell(\hat{\lambda})$ with $\hat{\lambda}_{\rho_i} \geq 3$.

In this case, removing c_i leaves row ρ_i with length $\hat{\lambda}_{\rho_i} - 1 \geq 2$, so the row remains in $\hat{\mu}_i$. Hence $\ell(\hat{\mu}_i) = \ell(\hat{\lambda})$ and $|\hat{\mu}_i| = |\hat{\lambda}| - 1$. Removing c_* removes one trailing 1-row, so $q(\mu_i) = q - 1$.

Then

$$q_0(\hat{\mu}_i) = |\hat{\mu}_i| - 2\ell(\hat{\mu}_i) + 2 = (|\hat{\lambda}| - 1) - 2\ell(\hat{\lambda}) + 2 = q_0(\hat{\lambda}) - 1,$$

and $q(\mu_i) = q - 1 \geq q_0(\hat{\lambda}) - 1 = q_0(\hat{\mu}_i)$. ✓

Case 2: $\rho_i = \ell(\hat{\lambda})$ and $\hat{\lambda}_{\rho_i} = 2$.

Removing $c_i = (\ell(\hat{\lambda}), 2)$ leaves row $\ell(\hat{\lambda})$ of length 1, so this row drops out of $\hat{\mu}_i$ (which records rows of length ≥ 2). $\hat{\mu}_i = \hat{\lambda}$ with the entire last row removed: $|\hat{\mu}_i| = |\hat{\lambda}| - 2$ and $\ell(\hat{\mu}_i) = \ell(\hat{\lambda}) - 1$.

What about $q(\mu_i)$? Counting trailing 1-rows of μ_i : removing c_* from λ kills one trailing 1-row, but the reclassified former- $\hat{\lambda}$ -last-row of length 1 becomes a new trailing 1-row. Net change: $q(\mu_i) = q$.

(Sanity check: $|\mu_i| = |\hat{\mu}_i| + q(\mu_i) = (|\hat{\lambda}| - 2) + q = |\lambda| - 2$. ✓)

Then

$$q_0(\hat{\mu}_i) = |\hat{\mu}_i| - 2\ell(\hat{\mu}_i) + 2 = (|\hat{\lambda}| - 2) - 2(\ell(\hat{\lambda}) - 1) + 2 = |\hat{\lambda}| - 2\ell(\hat{\lambda}) + 2 = q_0(\hat{\lambda}),$$

and $q(\mu_i) = q \geq q_0(\hat{\lambda}) = q_0(\hat{\mu}_i)$. ✓

In both cases, $q(\mu_i) \geq q_0(\hat{\mu}_i)$. □

Remark 7 (Dichotomy on corners). The case dichotomy $\rho_i = \ell(\hat{\lambda})$ vs. $\rho_i < \ell(\hat{\lambda})$ is exhaustive for length-preserving corners c_i of λ : if $\rho_i < \ell(\hat{\lambda})$, $\hat{\lambda}_{\rho_i} \geq 3$ automatically. (If $\hat{\lambda}_{\rho_i} = 2$ with $\rho_i < \ell(\hat{\lambda})$, then $\hat{\lambda}_{\rho_i+1} \leq 2$, but $\hat{\lambda}$ has all rows ≥ 2 , so $\hat{\lambda}_{\rho_i+1} = 2 = \hat{\lambda}_{\rho_i}$, making c_i *not* a corner — the row below has the same length.)

5 Main theorem

Theorem 8 (Main: $T_1 = -1$ on B in the deep stable regime). *Let $\lambda = (\hat{\lambda}, 1^q)$ with $\lambda_\ell = 1$ (i.e., $q \geq 1$). Assume:*

(a) *The SRD-style reduction (2) holds at λ and at every shape encountered in the recursive descent of λ .*

(b) $q \geq q_0(\hat{\lambda}) := |\hat{\lambda}| - 2\ell(\hat{\lambda}) + 2$.

Then

$$B_{\ell+1}^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda}.$$

Proof. We argue by induction on $|\hat{\lambda}^{(\geq 2)}| = |\hat{\lambda}| - \ell(\hat{\lambda})$.

Base case $|\hat{\lambda}^{(\geq 2)}| = 0$. Then $\hat{\lambda}$ has all rows of length 1, but our convention says $\hat{\lambda}$ consists of rows of length ≥ 2 . So $\hat{\lambda} = \emptyset$ and $\lambda = (1^q)$. By Lemma 4, $B^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda}$.

Inductive step. Suppose $|\hat{\lambda}^{(\geq 2)}| \geq 1$, so $\hat{\lambda} \neq \emptyset$ and λ has $r \geq 1$ length-preserving corners. By Lemma 5, hypothesis (b) implies $\ell(\lambda) \geq 3$. By Lemma 6, each μ_i satisfies its own threshold:

$q(\mu_i) \geq q_0(\hat{\mu}_i)$. Hypothesis (a) propagates to μ_i by definition of the fully stable regime: the recursive descent of μ_i is a subtree of that of λ , and SRD reduction holds throughout. Hence the inductive hypothesis applies to each μ_i :

$$B^{(\mu_i)}V_{\mu_i} \subseteq E_1^-|_{V_{\mu_i}}.$$

Apply Theorem 3 (using $\ell(\lambda) \geq 3$ and (2) at λ): $B^{(\lambda)}V_\lambda \subseteq E_1^-|_{V_\lambda}$.

Induction terminates because $|\hat{\mu}_i^{(\geq 2)}| < |\hat{\lambda}^{(\geq 2)}|$. Concretely, $|\hat{\mu}_i^{(\geq 2)}| = |\hat{\mu}_i| - \ell(\hat{\mu}_i)$.

- Case 1: $|\hat{\mu}_i^{(\geq 2)}| = (|\hat{\lambda}| - 1) - \ell(\hat{\lambda}) = |\hat{\lambda}^{(\geq 2)}| - 1$.
- Case 2: $|\hat{\mu}_i^{(\geq 2)}| = (|\hat{\lambda}| - 2) - (\ell(\hat{\lambda}) - 1) = |\hat{\lambda}^{(\geq 2)}| - 1$.

Both cases drop $|\hat{\lambda}^{(\geq 2)}|$ by exactly 1. □

Remark 9 (Pairing with the structural skeleton). Theorem 8 is dual to Theorem A of 2026-05-13-evening-T-ell which proved that T_ℓ acts as the scalar $+q$ on $B^{(\lambda)}V_\lambda$ for every λ with $\lambda_\ell = 1$ (no threshold required). The combined picture in the deep stable regime ($q \geq q_0(\hat{\lambda})$ and fully stable):

$$B^{(\lambda)}V_\lambda \subseteq E_\ell^+|_{V_\lambda} \cap E_1^-|_{V_\lambda}.$$

The T_ℓ -action is universal; the T_1 -action has a shape-dependent threshold. Both belong to the centraliser $Z_{H_q}(T_1) \cap Z_{H_q}(T_\ell)$ on B (and any operator preserving B as a subspace must commute with these two scalar actions, hence lies in the appropriate centraliser).

6 Threshold sharpness: computational evidence

We computed $T_1|_B$ for a range of test shapes via mod- P arithmetic ($P = 100\,003$, $q = 11$). For each shape $\lambda = (\hat{\lambda}, 1^q)$, we extracted a basis W of $B^{(\lambda)}V_\lambda$ and tested whether $T_1W = -W$ (i.e., $B \subseteq E_1^-$).

$\hat{\lambda}$	$q_0(\hat{\lambda})$	first q giving $B \subseteq E_1^-$	verdict
(2)	2	2	matches
(3)	3	3	matches
(4)	4	4	matches
(5)	5	5	matches
(2, 2)	2	2	matches
(3, 2)	3	3	matches
(3, 3)	4	4	matches
(4, 2)	4	4	matches
(4, 3)	5	$\geq 5^*$	matches at $q = 4$ (below)
(2, 2, 2)	2	2	matches

In every test, $q_0(\hat{\lambda})$ matches the empirical sharp threshold. The cells of the test grid below threshold show “ T_1 mixed” (neither $+q$ nor -1 on all of B); the cells at or above threshold show $T_1|_B = -1$.

* **Note on** $(4, 3)$. The cell $\hat{\lambda} = (4, 3)$ was verified *below* threshold ($q = 4$, $\dim V_{\lambda} = 1100$): $\dim B = 5$ (matches the dim closed form) but T_1 has mixed action on B — consistent with the prediction $q < q_0 = 5$. The above-threshold case $(4, 3, 1^5)$ ($n = 12$, $\dim V_{\lambda} = 4158$) exceeds the computational budget of the mod- P verification harness in this session and is left as a future check. The shape illustrates the precise distinction between *dim* stability and $T_1 = -1$ stability: the closed-form dim matches at $q = 4$ but T_1 -stability requires the strictly stronger condition $q \geq 5$.

Scripts. `~/projects/scratch/2026-05-13-night-j-formula/verify-T1_neg1.py.`

7 Honest scope

What is proved. Theorem 8 (inductive step + threshold propagation + column-shape base case). All three are short structural lemmas relying only on the index-gap commutation $|T_j, T_1| = 0$ for $j \geq 3$, the T_1 -equivariance of Φ_i^{λ} (for $n \geq 4$), and the SRD-style reduction hypothesis.

What is conditional. Hypothesis (a) (full stability of the SRD-style reduction along the descent) remains the substantive structural input. It has been proved in individual shape families (hooks, $(2^a, 1^?)$, $(3, 3, 1^?)$ at $q \geq 2$, $(4, 2, 1^?)$ at $q \geq 2$, $(4, 3, 1^?)$ at $q \geq 3$, and the $(3, 2, 1^?)$ family at all q) but not yet in full generality. A general structural proof of SRD-style reduction would render Theorem 8 unconditional in every fully stable shape.

Sharpness. The threshold $q_0(\hat{\lambda})$ matches the empirical sharp threshold on every shape tested in §6. The present paper proves it is *sufficient*; sharpness (i.e., that $B \not\subseteq E_1^-$ for $q < q_0$) is empirical only — a proof of necessity is a natural next target. Conceptually, a Case-2 step “borrows” a 1-row from $\hat{\lambda}$ to $\text{trail}(\lambda)$, keeping q_0 the same. A Case-1 step decrements both q and q_0 by 1 each. The threshold counts the worst-case descent depth needed for the chain $\mathcal{O}_{\lambda}$ at every level to have index ≥ 3 .

Relation to the morning refutation paper. The morning paper (2026-05-13-decoration-iso-failed.tex) observed $T_1|_B = -1$ on the test case $\lambda = (3, 2, 1^3)$ and on “4 other test cases” but did not extract a threshold. The threshold $q_0(\hat{\lambda}) = |\hat{\lambda}| - 2\ell(\hat{\lambda}) + 2$ identified here closes that observation into a clean structural statement.

What this enables. Combined with the evening structural-skeleton paper, we now have a two-sided structural skeleton for $B^{(\lambda)}V_{\lambda}$ in the deep stable regime:

1. T_{ℓ} acts as $+q$ (universal, evening paper).
2. T_1 acts as -1 (deep stable regime, this paper).
3. $E_1^-|_{V_{\lambda}} \subseteq \ker \Omega$, with $\Omega := R'_{\ell+1} \cdots R'_n$, so $\overline{\Omega} : V_{\lambda}/E_1^-|_{V_{\lambda}} \rightarrow B$ is surjective by construction (evening paper, Theorem A').

The combined picture gives strong restrictions on any potential Hecke-module structure on B : every operator preserving B as a subspace must commute with T_1 and T_{ℓ} , hence lies in $Z_{H_q}(T_1) \cap Z_{H_q}(T_{\ell})$. This is the natural setting in which to look for the elusive “decoration functor” upgrading the SYT-count equality to a representation-theoretic isomorphism.

Open problems.

- Prove the SRD-style reduction (2) structurally in full generality.
- Prove the threshold $q_0(\hat{\lambda})$ is sharp: $B \not\subseteq E_1^-$ for $q < q_0(\hat{\lambda})$.
- Describe $B^{(\lambda)}V_\lambda$ for $q < q_0(\hat{\lambda})$ — what is the T_1 -eigenspace decomposition of B on the non-stable side?